MILITARY SPECIFICATION

PERISCOPE: XM39

1. SCOPE

1.1 This specification covers one type of tank commander's binocular periscope known as the XM39. The periscope is a part of the fire control system in the 105mm Gun, Tank, M60 series, and is used for observation and machine gun sighting.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-O-13830	Optical Components for Fire Control Instruments; General Specification Governing the Manufacture, Assembly, and Inspection of
MIL-F-13926	Fire Control Materiel, General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Ordnance Materiel, Packaging of
MIL-F-14619	Fire Control Parts, Equipment and Tools; Packaging of
MIL-I-45208	Inspection Requirements, General Specification for
MIL-I-45607	Inspection Equipment, Supply and Maintenance for Ordnance

FSC 1240

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:42Z
Check the source to verify that this is the current version before use.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-130	Identification Marking of U. S. Property

DRAWINGS

Ordnance Corps

F8620610	Periscope, XM39
LSQAP 8620610	Supplementary Quality Assurance Provisions List for Periscope, XM39
IEL8620610	Inspection Equipment List for Periscope, XM39

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Initial production sample. - Unless otherwise directed by the contracting officer, an initial production sample shall be required. The sample shall be manufactured in the same manner, using the same materials, equipment, processes and procedures as will be used in regular production.

3.2 Materials. - Materials shall be in accordance with drawings, material specifications and general specifications forming a part of this specification.

3.3 Construction. - The periscope shall be manufactured in accordance with Drawing F8620610, "Periscope, XM39", and all drawings pertaining thereto. The requirements of this specification are detailed only to the extent considered necessary to obtain the desired optical and mechanical characteristics, performance and permanence.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:42Z
Check the source to verify that this is the current version before use.

3.4 General specifications. - The contractor shall be responsible for adherence to, and compliance with, the following requirements of Specifications MIL-O-13830 and MIL-F-13926:

3.4.1 MIL-O-13830:

(a) Cleanliness.

(b) Illuminated reticle.

3.4.2 MIL-F-13926:

(a) Order or precedence.

(b) Protective finished surfaces.

(c) Special accuracy.

(d) Finish, painting and protective.

(e) Workmanship.

3.5 Environmental.

3.5.1 Storage temperature. - The periscope shall show no evidence of cement separation, glass breakage or other physical failure and shall meet all the requirements of this specification at standard ambient temperature (plus 60 degrees to 90 degrees Fahrenheit) after having been exposed and thermally stabilized at ambient temperatures of minus 80 degrees and plus 160 degrees Fahrenheit.

3.5.2 Operating temperatures. - The periscope shall meet the applicable requirements of 3.8.1.2.2, 3.8.2.4.5, 3.8.2.5.2 and 3.8.2.7.1, while exposed and thermally stabilized at ambient temperatures of minus 40 degrees and plus 150 degrees Fahrenheit, subsequent to which the periscope shall meet the requirements of 3.5.3 at standard ambient temperature.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:42Z
Check the source to verify that this is the current version before use.

3.5.3 Vibration. - The periscope shall be vibrated at a constant frequency of 30 cycles per second with an amplitude of 1/16 inch (1/8 inch total excursion) for a period of 5 minutes plus or minus 15 seconds. Subsequent to vibration, there shall be no evidence of physical failure and the periscope shall meet the requirements of 3.6 to 3.10 inclusive.

3.6 Cleanliness. - The periscope shall meet the cleanliness requirements of Specification MIL-O-13830. When viewed through the eyepiece, the periscope shall have no more than 2 particles of dirt visible in the seven-power system. There shall be no dirt on the surfaces which interferes with observation or adversely affects image quality.

3.7 Sealing. - The periscope head and body assembly, when tested separately, shall show no evidence of leakage when subjected to an internal pressure of 5 psi for a period of 5 minutes, after having been exposed to the requirements of 3.5.2. The interior atmosphere of the accepted head and body assembly shall contain nitrogen having a dew point at least as low as minus 25 degrees Fahrenheit at atmospheric pressure.

3.8 Performance.

3.8.1 Head assembly. - With the head assembly securely mounted with datum plane "C" parallel with a plumb line within 10 seconds, and with datum plane "A" making an angle of 20 degrees, plus or minus 10 seconds, with the plumb line (datum planes shown on Drawing F8620599) and with the head mirror at an angle of 45 degrees from the horizontal, the head assembly shall meet the following requirements.

3.8.1.1 Collimation.

3.8.1.1.1 Deflection. - A beam of light perpendicular to datum plane "C" (Drawing F8620599) within 0.1 mil shall be reflected 90 degrees plus or minus 0.2 mil.

3.8.1.1.2 Resolution. - The resolving power shall be within 6 seconds of arc or less throughout the entire aperture when viewed at zero degrees elevation with a twenty-power telescope with a 1-1/2 inch aperture.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:42Z
Check the source to verify that this is the current version before use.

3.8.1.2 Elevation.

3.8.1.2.1 Elevation input shaft travel. - Rotation of the elevation input shall shall be limited by contact with the travel stop pins only. The shaft shall rotate a minimum of 60 degrees elevation and 15 degrees depression. (See 6.5)

3.8.1.2.2 Torque. - The running torque required to rotate the elevation input shaft through maximum elevation and depression shall not exceed 10 pound-inches at standard ambient temperature or 13 pound-inches at extreme temperatures of minus 40 degrees and plus 150 degrees Fahrenheit. (See 6.5)

3.8.1.2.3 Plumb travel. - After establishing coincidence with a plumb line at zero degrees elevation, the line of sight shall not deviate from the plumb line by more than 0.20 mil, nor shall the total lateral spread exceed 0.20 mil, when the line of sight is elevated 15 degrees and depressed 10 degrees by means of the elevation input arm. The line of sight shall not deviate from the plumb line by more than 0.5 mil from 15 to 60 degrees elevation and 10 to 15 degrees depression.

3.8.1.2.4 Elevation accuracy. - The angle between the line of sight through the reticle boresight mark and the line through the centers of the two bores of the elevation input arm shall remain constant within plus or minus 0.2 mil, excluding backlash, at any setting of the arm between 10 degrees depression and 20 degrees elevation, and within plus or minus 1.0 mil at any setting above 20 degrees elevation up to and including 60 degrees, and from 10 degrees depression down to and including 15 degrees depression. The elevation or depression angle is defined as the angle between the line of sight and the horizontal datum plane of the instrument.

3.8.1.2.5 Backlash. - Backlash between the optical line of sight and the elevation input arm shall not exceed 0.2 mil at any setting of the arm.

3.8.2 Body assembly.

3.8.2.1 Collimation. - When two parallel incident rays perpendicular to the mounting surface pass through the binocular body, the conjugate rays emerging from the two eyepieces shall be parallel within the values given in 3.8.2.1.1 and 3.8.2.1.2 at infinity focus of the eyepiece and for any setting of the interpupillary distance scale between 58 and 72 millimeters (mm).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:42Z
Check the source to verify that this is the current version before use.

3.8.2.1.1 Collimation, vertical plane. - Dipvergence shall not exceed 14 minutes of arc (2.0 minutes of true field) in the direction perpendicular to the plane determined by the two entrant rays.

3.8.2.1.2 Collimation, horizontal plane. - Divergence shall not exceed 28 minutes of arc (4.0 minutes of true field), nor shall convergence exceed 14 minutes of arc (2.0 minutes of true field).

3.8.2.2 Reticle plumb. - With the head assembly positioned as described in 3.8.1, the 400 meter horizontal reticle line shall be parallel to plane "B" within plus or minus 24 minutes.

3.8.2.3 Parallelism. - Binocular images shall be parallel within 30 minutes.

3.8.2.3.1 Parallelism of image and reticle. - With the periscope mounted within 10 seconds of square vertically, the image of a horizontal line shall be parallel to the horizontal reticle line within 30 minutes of arc.

3.8.2.4 Boresight knobs.

3.8.2.4.1 Elevation knob travel. - The boresight knob shall provide a minimum travel of 5.5 mils in elevation and 8.5 mils in depression from the collimated setting of 3.8.2.1.

3.8.2.4.2 Deflection knob travel. - The boresight knob shall provide a minimum travel of 7 mils in each direction from the collimated setting of 3.8.2.1.

3.8.2.4.3 Reticle shift. - Rotation of the boresight knobs shall not cause the reticle to shift more than 0.1 mil in a direction perpendicular to the movement of the reticle cell.

3.8.2.4.4 Backlash. - Backlash in each boresight mechanism shall not exceed 0.10 mil.

3.8.2.4.5 Disengagement of boresight clutch and knobs. - The force required to disengage the boresight knob clutch shall not exceed 5 pounds at standard ambient temperature or 8 pounds at extreme temperatures as specified in 3.5.2. The clutch, when disengaged, shall allow the knobs to rotate in each direction within a running torque of not more than 5 pound-inches at standard ambient temperature or 8 pound-inches at extreme temperatures as specified in 3.5.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:42Z
Check the source to verify that this is the current version before use.

3.8.2.5 Eyepiece focus. - The diopter scale index shall indicate zero within plus or minus 1/4 diopter when the reticle is brought into sharp focus with the aid of a dioptometer of at least 3 power or an equivalent auxiliary telescope. From this sharp focus, the eyepiece shall have at least 3.0 diopter movement in both the plus and minus direction.

3.8.2.5.1 Eyepiece alinement. - With the same diopter setting on each eyepiece, one eyepiece shall not extend beyond the other by more than 0.05 inch.

3.8.2.5.2 Diopter adjustment torque. - The running torque required to rotate the diopter adjustment in both body assemblies shall not exceed 7 pound-inches at standard ambient temperature or 10 pound-inches at extreme temperatures as specified in 3.5.2.

3.8.2.5.3 Diopter backlash. - Backlash in the diopter movement shall not exceed 1/8 diopter.

3.8.2.6 Reticle illumination. - With the reticle light energized with a minimum of 18 volts and maximum of 30 volts, the reticle marking shall appear clearly defined when observed in a dark room.

3.8.2.7 Interpupillary distance (IPD). - The IPD scale shall be correct to within plus or minus 1 millimeter of actual distance between optical axis of eyepieces.

3.8.2.7.1 Interpupillary distance knob torque. - The IPD knob torque shall not exceed 7 pound-inches at standard ambient temperature or 10 pound-inches at extreme temperatures as specified in 3.5.2.

3.8.2.8 Resolution. - The resolving power shall be 6 seconds of arc or less on the axis for a parallel bar test.

3.8.3 Complete periscope.

3.8.3.1 Repeatability. - The line of sight of the periscope shall not deviate from an original setting by more than 0.2 mil in elevation or deflection when the body assembly is removed and reattached to the head assembly.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:42Z
Check the source to verify that this is the current version before use.

3.8.3.2 Line of sight retention. - The line of sight of the periscope shall be retained within plus or minus 0.2 mil in elevation and deflection from an original setting subsequent to the vibration specified in 3.5.3.

3.9 Parallax. - Parallax at the optical axis of the seven-power binocular system shall not exceed 0.2 mil when viewing a target from 500 meters to infinity.

3.10 Identification marking. - Identification marking, unless otherwise specified, shall be in accordance with Standard MIL-STD-130.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibilities. - The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conforming to prescribed requirements.

4.2 General provisions. - Inspection requirements and procedures shall be in accordance with Specification MIL-I-45208.

4.3 Initial production sample. - Prior to production, the contractor shall submit the initial production sample in accordance with 3.1 for inspection by the contractor under the surveillance of the Government. This inspection shall be conducted at the contractor's facility, or Government approved laboratory, using test equipment authorized for use on the contract. The sample shall be inspected for all requirements of this specification by utilizing examination and testing methods prescribed herein.

4.3.1 Production sample failure. - Should the production sample fail to pass the prescribed tests, the contractor shall cease those operations which are causing the rejection and shall make the necessary corrections. Where corrections in the process have been made, a new production sample shall be examined and tested by the contractor under the surveillance of the Government. Acceptance of production lots will not commence until an acceptable production sample is submitted and corrective actions have been taken to eliminate deficiencies.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:42Z
Check the source to verify that this is the current version before use.

4.4 Lot formation. - Inspection lot sizes, lot formation, and sampling for inspection shall be in accordance with Standard MIL-STD-105.

4.4.1 Defective units and lots shall be processed in accordance with Standard MIL-STD-105.

4.5 Material. - Control of material shall be in accordance with test data requirements of Specification MIL-I-45208.

4.6 Inspection of components. - Inspection requirements for component parts and subassemblies are contained herein and on Supplementary Quality Assurance Provisions (SQAP) listed on Supplementary Quality Assurance Provisions List (LSQAP) 8620610.

4.6.1 Contents of SQAP. - The contents of SQAP shall represent the minimum inspection requirements and shall be the basis for acceptance or rejection by lot of those components covered by a SQAP. The Government reserves the right, however, to inspect for the requirements of all drawings and specifications and to reject individual non-conforming items.

4.6.2 Contractor's inspection for all components not covered by SQAP. - Inspection of these components shall be in accordance with the inspection procedures of the written inspection system required by Specification MIL-I-45208. The system shall include procedures for checking all positional and functional characteristics shown on the drawings which indicate precision requirements and have a bearing on the functional accuracy and performance of the periscope.

4.7 Inspection equipment. - The inspection equipment shall conform to that listed in IEL-8620610 except as provided for under the alternate inspection procedures and inspection equipment requirements of Specification MIL-I-45208. The provisions of Specification MIL-I-45607 shall apply to the test equipment provided by the Government or provided by the contractor and utilized in the Government approved inspection system. The Government reserves the right to use all approved test equipment, both Government furnished and contractor acquired, for verification inspection.

4.7.1 Accuracy requirements. - Inspection equipment which incorporates features for measuring values or sizes established by product requirements shall be capable of measuring those values within an accuracy of 10 percent of the specified tolerance, except as otherwise specified. In cases where the 10 percent

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:42Z
Check the source to verify that this is the current version before use.

accuracy imposes impractical or very difficult design and fabrication problems, or results in a test equipment cost factor which is exorbitant in relation to the cost and application of the product to be inspected, the Government shall be expeditiously advised in order that necessary appropriate action can be taken.

4.7.2 Inspection equipment design requirements.

4.7.2.1 Equipment to test vibration requirements. - The equipment shall be a type capable of producing vertical motion at a constant frequency of 30 cycles per second with an amplitude of 1/16 inch (1/8 inch total excursion). The equipment shall be equipped with an adapter which will permit the mounting of each complete periscope in its normal using position with the input coupling immobilized. The base of the adapter shall be secured firmly to the device.

4.7.2.2 Equipment to test head assembly collimation, plumb travel, and elevation accuracy. - The equipment shall permit mounting the head assembly as specified in 3.8.1 and shall meet the following requirements:

(a) Collimation. - The medium used for testing collimation shall be accurate within 10 seconds throughout the seven-power viewing area of the device.

(b) Plumb travel. - The medium used to test plumb travel shall be perpendicular within 10 seconds with respect to the plumb line plane specified in 3.8.1.

(c) Elevation accuracy. - The input coupling device may compensate, both in the vertical and horizontal plane, but the mounting surface shall remain stable within 10 seconds to the arrested zero line of sight position; also throughout the entire excursion of rotation required for testing angular accuracy.

(d) The device shall be capable of positioning the input coupling to any measured angular position throughout its entire excursion.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:42Z
Check the source to verify that this is the current version before use.

4.7.2.3 Equipment to test head assembly torque requirements (input coupling). - A torque wrench type tester with a maximum range of 10 pound-inches, the accuracy of which shall be plus or minus 1/4 pound-inch throughout the entire zone of excursion, shall be utilized. This device shall be equipped with a suitable adapter to accommodate testing the input coupling.

4.7.2.4 Equipment to test body assembly collimation, IPD, reticle plumb, reticle illumination, parallelism, and parallelism of reticle and image. - The equipment shall permit mounting the body assembly as specified in 3.8.2.2. The device shall also be equipped with a 24 volt variable power source for energizing the reticle light and shall meet the following requirements:

(a) Collimation. - The medium used for testing vertical and horizontal collimation shall be accurate within 10 seconds throughout the seven-power viewing area of the device at the interpupillary distances specified in 3.8.2.1.

(b) Reticle shift. - The medium used for testing reticle shift shall be accurate within 10 seconds.

(c) Reticle plumb. - The medium used to test line of sight relationship shall be accurate within 10 seconds.

(d) Backlash. - The medium used for testing backlash shall be accurate within 10 seconds.

(e) Parallelism. - An auxiliary telescope with appropriate targets shall be used, capable of adjustment at either body assembly eyepiece and determining image parallelism within 3 minutes.

(f) Parallelism of image and reticle. - The medium used for testing parallelism of image and reticle shall be perpendicular respectively within 10 seconds to the plane specified in 3.8.2.3.1.

4.7.2.5 Equipment to test assembly torque requirements. - A torque wrench type tester with a maximum range of 15 to 30 pound-inches, the accuracy of which shall be plus or minus 1/4 pound-inch throughout the entire zone of

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:42Z
Check the source to verify that this is the current version before use.

excursion, shall be utilized. This equipment shall be equipped with suitable adapters to accommodate testing boresight knobs, diopter adjustment and IPD knobs.

4.7.2.6 Equipment to test body assembly direct force requirements (disengagement of boresight clutch). - An instrument used to measure direct force, push or pull, in pounds, with a range of at least 10 pounds and an accuracy of plus or minus 2 ounces throughout the entire range, shall be utilized. This equipment shall be equipped with a suitable adapter to accommodate testing the disengagement of the boresight clutch.

4.7.2.7 Equipment to test sealing. - The equipment to test sealing of the body assembly or head assembly shall permit nitrogen of specified pressure and dew point to be inserted into all sections of the assemblies and to hold the pressure for the time specified in 3.7. The equipment shall be capable of reading pressure drop in .05 psi increments, and dew point at least as low as minus 25 degrees Fahrenheit.

4.7.2.8 Equipment to test eyepiece focus. - This equipment shall be a dioptometer having a magnification of at least three power and graduations of 1/8 diopter or better.

4.8 Acceptance inspection. - Classification of defects in Tables I, II, III and IV shall constitute the contractor's minimum inspection to be performed for acceptance or rejection by lots on a characteristic basis.

Table I

Classification of Defects

Use Inspection Level II in Table IIIA with Sampling Plan Table IVA of Standard MIL-STD-105.

Critical: None defined.

Major: AQL 1.5 Percent Defective	Req't. Par.	Q.A. Par.	Test Method
101. Completeness of assembly	3.3	--	Visual
102. Cleanliness	3.6	4.9.2	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:42Z
Check the source to verify that this is the current version before use.

Table I - Cont'd.

Major: AQL 1.5 Percent Defective	Req't Par.	Q.A. Par.	Test Method
103. Parallax	3.9	4.9.6	Visual - Special Testing Equipment (STE)
104. Identification marking	3.10	4.9.7	Visual

Minor: None defined.

NOTE: The tests of Table I shall be performed at standard ambient temperature (plus 60 degrees to 90 degrees Fahrenheit).

Table II

Classification of Defects for Head Assembly

Use Inspection Level II in Table IIIA with Sampling Plan Table IVA of Standard MIL-STD-105

Critical: None defined.

Major: AQL 1.5 Percent Defective	Req't Par.	Q.A. Par.	Test Method
101. Mechanical orientation	3.8.1	4.9.4	Visual - STE
102. Deflection	3.8.1.1.1	4.9.4.1	Visual - STE
103. Resolution	3.8.1.1.2	4.9.4.2	Visual - STE
104. Elevation input shaft travel	3.8.1.2.1	4.9.4.3	Manual - Visual
105. Elevation input shaft torque	3.8.1.2.2	4.9.4.4	Visual - STE
106. Plumb travel	3.8.1.2.3	4.9.4.5	Visual - STE
107. Elevation accuracy	3.8.1.2.4	4.9.4.6	Visual - STE
108. Backlash	3.8.1.2.5	4.9.4.7	Visual - STE

Minor: None defined.

NOTE: The tests of Table II shall be performed at standard ambient temperature (plus 60 degrees to 90 degrees Fahrenheit).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:42Z
Check the source to verify that this is the current version before use.

Table III

Classification of Defects for Body Assembly

Use Inspection Level II in Table IIIA with Sampling Plan Table IVA of Standard MIL-STD-105

Critical: None defined.

Major: AQL 1.5 Percent Defective	Req't Par.	Q.A. Par.	Test Method
101. Collimation, vertical	3.8.2.1.1	4.9.5.1	Visual - STE
102. Collimation, horizontal	3.8.2.1.2	4.9.5.2	Visual - STE
103. Reticle plumb	3.8.2.2	4.9.5.3	Visual - STE
104. Binocular image parallel	3.8.2.3	4.9.5.4	Visual - STE
105. Parallelism of image and reticle	3.8.2.3.1	4.9.5.5	Visual - STE
106. Boresight knob travel, elevation	3.8.2.4.1	4.9.5.6	Visual - STE
107. Boresight knob travel, deflection	3.8.2.4.2	4.9.5.7	Visual - STE
108. Reticle shift	3.8.2.4.3	4.9.5.8	Visual - STE
109. Boresight knob backlash	3.8.2.4.4	4.9.5.9	Visual - STE
110. Boresight knob torque	3.8.2.4.5	4.9.5.10	Visual - STE
111. Eyepiece focus	3.8.2.5	4.9.5.12	Visual - STE
112. Eyepiece alinement	3.8.2.5.1	4.9.5.13	Standard Measuring Equipment (SME)
113. Diopter adjustment torque	3.8.2.5.2	4.9.5.10	Visual - STE
114. Diopter movement backlash	3.8.2.5.3	4.9.5.14	Visual - STE
115. Reticle illumination	3.8.2.6	4.9.5.15	Visual - STE
116. IPD scale accuracy	3.8.2.7	4.9.5.16	Visual - STE
117. IPD knob torque	3.8.2.7.1	4.9.5.10	Visual - STE
118. Resolution	3.8.2.8	4.9.5.17	Visual - STE

Minor: None defined.

NOTE: The tests of Table III shall be performed at standard ambient temperature (plus 60 degrees to 90 degrees Fahrenheit).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:42Z
Check the source to verify that this is the current version before use.

Table IV

Inspection Testing

Use Inspection Level L7 in Table IIIB with Sampling Plan Table IVA of Standard MIL-STD-105

Critical: None defined.

Major: AQL 6.5 Percent Defective	Req't Par.	Q.A. Par.	Test Method
101. IPD knob torque	3.5.2	4.9.1.1 4.9.1.3	STE
102. Boresight knob torque	3.5.2	4.9.1.1 4.9.1.3	STE
103. Elevation input shaft torque	3.5.2	4.9.1.1 4.9.1.3	STE
104. Boresight clutch force	3.5.2	4.9.1.1 4.9.1.3	STE
105. Diopter adjustment torque	3.5.2	4.9.1.1 4.9.1.3	STE
106. Sealing (head only)	3.7	4.9.3	STE
107. Sealing (body only)	3.7	4.9.3	STE
108. Vibration	3.5.3	4.9.1.4	STE
109. Repeatability	3.8.3.1	4.9.8	STE
110. Line of sight retention	3.8.3.2	4.9.9	STE

Minor: None defined.

NOTE: The tests for characteristics 101, 102, 103, 104 and 105 shall be conducted while the instrument is stabilized at temperatures of minus 40 degrees and plus 150 degrees Fahrenheit. The tests for characteristics 106, 107, 108, 109, and 110 shall be conducted at standard ambient temperature (plus 60 degrees to 90 degrees Fahrenheit). All tests shall be conducted in accordance with the Quality Assurance Paragraph (Q.A. Par.) listed.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:42Z
Check the source to verify that this is the current version before use.

4.9 Test methods and procedures.

4.9.1 Environmental.

4.9.1.1 Pre-environmental torque tests. - The torque and force required to rotate or disengage the input coupling, boresight clutch, boresight knobs, diopter adjustment, and IPD knob shall be measured and recorded at standard ambient temperature prior to environmental testing of the periscope. The test of the applicable control shall be conducted as specified in 4.9.4.4 and 4.9.5.10.

4.9.1.2 Storage temperature. - This test shall be conducted in accordance with procedure II of Specification MIL-F-13926 to the temperatures specified in 3.5.1. Upon completion of the specified exposure, the periscope shall be inspected for damage and then subjected to the tests specified in 4.9.2, 4.9.4, 4.9.5, and 4.9.6 and shall meet the requirements specified. This test shall be performed on the unit specified in 4.3 only.

4.9.1.3 Operating temperatures. - This test shall be conducted in accordance with procedure I of Specification MIL-F-13926 except that the temperature shall not be reduced beyond condition "C" or raised beyond condition "G". When the periscope is at the indicated temperature, it shall be subjected to the tests specified in 4.9.4.4 and 4.9.5.10. The controls of the periscope shall meet the specified torque and force requirements for the indicated operating temperatures as specified in 3.8.1.2.2.

4.9.1.4 Vibration. - The periscope shall be mounted in its normal operating position on a special testing device and vibrated in accordance with 3.5.3. During this test the input coupling shall be immobilized in any position except that the periscope shall not contact the travel stop pin. Subsequent to vibration, the periscope shall be subjected to a visual-tactile examination for damage. The periscope shall then be subjected to the tests specified in 4.9.2, 4.9.4, 4.9.5, and 4.9.6 and shall meet the requirements specified.

4.9.2 Cleanliness. - To determine conformance to 3.6, the visual systems of the periscope shall be inspected for cleanliness in accordance with Specification MIL-O-13830 and shall meet the requirements specified. The mean diameter of a size 10 dig is equal to 0.10 millimeter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:42Z
Check the source to verify that this is the current version before use.

4.9.3 Sealing. - The head assembly and body assembly shall be pressurized to the extent specified in 3.7 using special testing equipment. Upon completion of the test, the assemblies shall be bled to the pre-test pressure and sealed. This is a post environmental test that shall be performed at standard ambient temperature.

4.9.4 Performance, head assembly. - The following tests shall be conducted with the head assembly secured in a special testing device as specified in 4.7.

4.9.4.1 Deflection. - The test shall be performed with a special testing device that shall be capable of accurately measuring any error in the deviation of the reflected line of sight with the incident line of sight established as specified. Collimation of the head assembly shall meet the requirements specified in 3.8.1.1.1.

4.9.4.2 Resolution, the entire aperture. - This test shall be conducted in accordance with Specification MIL-O-13830. Resolution shall meet the requirements specified in 3.8.1.1.2.

4.9.4.3 Shaft travel. - The excursion of the input shaft shall be tested manually and shall meet the requirements specified in 3.8.1.2.1.

4.9.4.4 Torque, input coupling. - The torque required to rotate the input coupling through the specified excursion shall be measured with a special testing device affixed to the coupling. This device shall be capable of measuring applied torques. The measured torque shall not exceed the value specified in 3.8.1.2.2 for the temperature indicated.

4.9.4.5 Plumb travel. - Direct a line of sight through the viewing area of the head assembly to a calibrated plumb line target. The lines of sight shall not deviate more than the tolerance specified in 3.8.1.2.3 when the coupling is rotated through the excursion indicated.

4.9.4.6 Elevation accuracy. - Direct a line of sight through the viewing area of the head assembly to a calibrated target appropriate for measuring elevation accuracy of the line of sight. Any measured change in angular position of the input coupling shall cause the same measured change in the line of sight within the tolerance specified in 3.8.1.2.4. Elevation accuracy shall be measured at a minimum of four positions in elevation and three positions in depression starting with the line of sight horizontal. One of these positions shall be at 60 degrees elevation and another at 15 degrees depression.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:42Z
Check the source to verify that this is the current version before use.

4.9.4.7 Backlash. - This test shall be performed utilizing the equipment specified in 4.9.4.1. With backlash excluded for direction of travel, record position of the input coupling and line of sight. Increase the angular position of the coupling approximately 10 degrees; then, with backlash included, return the coupling to the recorded position without overtravel and record position of the line of sight. The difference between the two recorded positions of the line of sight is the amount of backlash. Backlash shall not exceed the tolerance specified in 3.8.1.2.5.

4.9.5 Performance, body assembly. - The following tests shall be conducted with the body assembly secured in a special testing device that shall position the assembly as specified in 3.8.2.2. The device shall permit a line of sight through the reticle boresight mark of the body assembly to be directed to a collimator type target. The target shall permit accurate measurement of the applicable characteristics of the body.

4.9.5.1 Collimation, vertical. - Dipvergence shall be tested by placing the binocular assembly on a special testing device that shall position the assembly as specified in 4.7.2.4. Parallel incident rays shall be passed through the body and observed at the eyepiece with a double collimator for deviation in the vertical plane. This test shall be performed with the IPD scale set at 58, 65, and 72 millimeters and shall meet the requirements specified in 3.8.2.1 and 3.8.2.1.1.

4.9.5.2 Collimation, horizontal. - Divergence and convergence shall be tested in conjunction with 4.9.5.1 by observing the horizontal rays at the eyepiece for conformance to the requirements specified in 3.8.2.1.2.

4.9.5.3 Reticle plumb. - The reticle line angle shall be examined by placing the body assembly on a special testing device that shall position the body as specified. The 400 meter horizontal reticle shall be projected through the exit portion of the body assembly and the angle formed by the line and the plumb line shall be within the tolerance specified in 3.8.2.2.

4.9.5.4 Binocular image parallelism. - With the aid of an auxiliary telescope and appropriate target, sight through one of the body assembly eyepieces and adjust the auxiliary telescope reticle to superimpose on the target. Translate the telescope to the other body assembly eyepiece and sight again at its target. The target images projected through the two eyepieces shall be parallel within the tolerances specified in 3.8.2.3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:42Z
Check the source to verify that this is the current version before use.

4.9.5.5 Parallelism of image and reticle. - When one extremity of the 400 meter horizontal reticle line is coincident to the image of a horizontal target line, the reticle line shall be parallel to the target line within the tolerance specified in 3.8.2.3.1 as measured on the target.

4.9.5.6 Boresight knob travel, elevation. - From the collimated position established in 3.8.2.1, rotate the knob to its clockwise stop and then to its counterclockwise stop. This knob shall move the reticle boresight point the distance specified in 3.8.2.4.1 as measured on the target.

4.9.5.7 Boresight knob travel, deflection. - Perform this test as specified in 4.9.5.6. Knob travel shall meet the requirements specified in 3.8.2.4.2.

4.9.5.8 Reticle shift. - This test shall be conducted concurrently with the tests specified in 4.9.5.6 and 4.9.5.7. During rotation of the knobs, the reticle boresight point shall not shift perpendicular to the line of travel more than the tolerance specified in 3.8.2.4.3.

4.9.5.9 Boresight knob backlash. - With backlash excluded for direction of travel, record position of knob and line of sight. Rotate the knob one-half of a revolution; then, with backlash included, return the knob to the recorded position without overtravel and record position of the line of sight. The difference between the two recorded positions of the line of sight is the amount of backlash. Backlash shall not exceed the tolerance specified in 3.8.2.4.4.

4.9.5.10 Torque, boresight knobs, IPD knob and diopter adjustment. - The torque required to rotate the boresight knobs, IPD knob and diopter adjustment shall be measured with special testing equipment affixed to the boresight knobs, IPD knob, and diopter adjustment. This equipment shall be capable of measuring applied torques. The measured torque of the boresight knobs, IPD knob and diopter adjustment shall not exceed the value specified in 3.8.2.4.5, 3.8.2.5.2, and 3.8.2.7.1 for the temperature indicated.

4.9.5.11 Disengagement of boresight clutch. - The force required to disengage the boresight clutch from the locked position shall be measured with a special testing device. This device shall be capable of measuring applied force. The measured force shall not exceed the value specified for the temperature indicated. Slip scales shall be manually tested at the temperatures indicated and shall meet the requirements specified in 3.8.2.4.5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:42Z
Check the source to verify that this is the current version before use.

4.9.5.12 Eyepiece focus. - This test shall be conducted with a dioptometer as specified in 4.7.2.8. The dioptometer shall be adjusted to its best focus and the diopter scale set to zero. Adjust the diopter scale of the body assembly to zero and observe the body assembly reticle through the dioptometer. After positioning the body reticle to clear focus, the scale shall be within the tolerance specified in 3.8.2.5. Proceed with the test by determining diopter adjustment range of the diopter scale.

4.9.5.13 Eyepiece alinement. - The eyepiece alinement shall be measured for conformance to the requirements specified in 3.8.2.5.1.

4.9.5.14 Diopter backlash. - The diopter movement shall be manually and visually tested for backlash. Backlash shall not exceed the tolerance specified in 3.8.2.5.3.

4.9.5.15 Reticle illumination. - This test shall be conducted in a dark area with the reticle lamp energized by a 24 volt source. The source shall contain a rheostat for varying reticle illumination for best definition. The reticle markings shall meet the requirements of 3.8.2.6.

4.9.5.16 Interpupillary distance (IPD) scale. - The IPD scale, when activated by its control knob, shall cover the limits specified in 3.8.2.7.

4.9.5.17 Body assembly resolution. - This test shall be conducted in accordance with Specification MIL-O-13830. Resolution shall meet the requirements specified in 3.8.2.8.

4.9.6 Parallax. - Position the periscope (head assembly and body assembly) in its normal operating position in a testing device. Adjust boresight knobs of a visual system to their midpoint of travel. Direct a line of sight through the periscope boresight point to a sharply defined target located at the specified distance from the periscope window. Parallax will be recognized as any apparent movement of a reticle in relation to the target image when the observer's head is moved from side to side and up and down. Parallax at the boresight point with respect to the target image shall not exceed the tolerance specified in 3.9.

4.9.7 Identification marking. - Identification marking, unless otherwise specified, shall be visually examined to determine conformance to Standard MIL-STD-130.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:42Z
Check the source to verify that this is the current version before use.

4.9.8 Repeatability. - To determine conformance to 3.8.3.1, the head assembly shall be mounted in its normal operating position and the body assembly shall be clamped, the line of sight on a convenient target recorded and the body assembly then unclamped. This procedure shall be repeated for four (4) complete cycles. The four determinations shall be averaged and the resulting value assigned as the line of sight deviation for the instrument.

4.9.9 Retention. - To determine conformance to 3.8.3.2, this test shall be performed in conjunction with the vibration requirements of 3.5.3. Prior to vibration the periscope, in its vibration fixture, shall be alined on a convenient target. The fixture location, with respect to the vibration medium, and the periscope location, with respect to the fixture, shall be noted. After vibration, the fixture shall be returned to its initial position and the periscope position verified with respect to the fixture. The line of sight shall be as specified in 3.8.3.2.

4.10 Quality assurance provisions for packaging, packing, and marking. - Quality assurance provisions for packaging, packing, and marking shall be in accordance with Specification MIL-P-14232.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking. - Packaging, packing, and marking shall be in accordance with Specification MIL-F-14619/P8620610. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. - The Periscope, XM39, is a binocular tank commander's periscope primarily intended for use in the fire control system of the 105mm Gun, Tank, M60 series, and is used for machine gun sighting and observing.

6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) Selection of applicable levels of preservation, packaging and packing.

(c) Applicable stock number.

(d) Provisions for production sample, including date of delivery. (See 3.1)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:42Z
Check the source to verify that this is the current version before use.

6.3 Design information. - Information regarding specific design not covered or referenced by this specification may be obtained from the procuring activity upon the special request of the contractor.

6.4 Definitions.

6.4.1 Tactile. - Skillful manipulation or sensing.

6.4.2 Standard measuring equipment (SME). - Common hand type measuring devices and standard gages which are usually stocked by commercial supply houses for ready supply (shelf items), and which are normally used by the inspector to perform dimensional inspection. This category also includes commercial testing equipment such as meters, optical comparators, etc.

6.4.3 Special testing equipment (STE). - Special inspection fixtures, instruments and modified standard measuring equipment and special purpose commercially available equipment.

6.4.4 Interpupillary distance (IPD). - The interpupillary distance is the distance between the optical axis of the two eyepieces.

6.5 Elevation shaft. - At no time shall the elevation shaft be allowed to strike hard against the travel stop screw.

NOTICE: When Government drawings, specifications or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any invention that may in any way be related thereto.

Custodian:

Army - MU

Preparing Activity:

Army - MU

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:42Z
Check the source to verify that this is the current version before use.